

MADNE GIRNARI

Data Analyst | Audit Analytics | ML & GenAI Solutions
Mumbai, India | +91 9987703145 | mmgmadne@gmail.com

PROFESSIONAL SUMMARY

Results-driven Senior Data Analyst with 8+ years of progressive experience in audit analytics, financial risk, and enterprise data engineering within leading banking institutions. Proven track record of deploying machine learning models, building executive-grade BI dashboards, and developing GenAI-powered automation tools that reduce manual effort and enhance compliance coverage. Deep expertise in SQL, Python, PySpark, and SAS across cloud-native environments (AWS, Cloudera/HDFS). Adept at translating complex financial data into actionable intelligence for audit, risk stakeholders at scale.

TECHNICAL SKILLS

Programming & Engineering: Python (Pandas, Scikit-Learn), PySpark, PL/SQL, Unix Shell Scripting

Machine Learning / AI: Logistic Regression, KNN, SVM, Anomaly Detection, Benford's Law, Levenshtein Algorithm, Neural Networks, Deep Learning, GPT-4o (GenAI)

Databases & Cloud: Oracle SQL 11g/12c, Impala, HDFS/Cloudera, AWS, Anaconda Enterprise

BI & Visualization: Power BI, SAS Viya, SAS Enterprise Guide, SAS Job Execution, Oracle Developer, Jupyter Notebook, Cloud Machine Learning

Domain Expertise: Internal Audit Analytics, Liquidity Risk (LCR/NSFR), Core Banking Systems (FinnOne LMS, TCS B@ncs24, Novopay, Credit Card Prime)

WORK EXPERIENCE

IDFC FIRST BANK | Mumbai, India

Aug 2023 – Present

Data Analyst – Audit Analytics & Centre of Excellence (COE)

- **Engineered STARK**, an enterprise GenAI utility powered by GPT-4o LLM, enabling automated SQL query generation from plain-language developer requirements; Project like **CBS ACM** to validate and audit employee access entitlements, ensuring correct access provisioning across banking applications. Automated **incentive retail asset calculation query generation** by orchestrating a rules-based engine across 10+ input source files, automatically producing structured target output files within the LLM workflow. Implemented **RAG with semantic chunking** and high-token chunking strategies to support diverse banking applications, enable dynamic chat flows, and significantly reduce LLM hallucinations — delivering audit-grade query accuracy and reliability at scale.
- **Engineered end-to-end ML pipelines** on Cloudera/HDFS using PySpark and SAS Viya to process millions of banking transactions daily, delivering complex trigger-based audit reports for the Internal Audit team with enhanced anomaly detection accuracy.
- **Deployed ensemble ML models** (Logistic Regression, KNN, SVM, Benford's Law, Levenshtein Algorithm) in Anaconda Enterprise to detect financial anomalies in audit data, validated with confusion matrix benchmarks achieving 85%+ classification accuracy.
- **Developed executive Power BI dashboards** monitoring 10+ operational and risk KPIs across audit findings, reducing manual reporting cycles by 40% and enabling real-time stakeholder visibility.
- **Developed and automated number of audit triggers** for the Internal Audit team using CML library like smtplib, Pandas, Logging enabling scheduled, concurrent execution of data validation checks on a monthly basis — eliminating manual query runs and improving audit exception detection efficiency.
- **Led Liquidity Risk analytics** for LCR and NSFR metrics, providing data-driven insights to senior risk management teams to ensure regulatory compliance with RBI frameworks.
- **Partnered with audit, risk, and technology stakeholders** across FinnOne LMS, TCS B@ncs24, Novopay, and Credit Card Prime data sources to define analytical requirements and deliver high-impact compliance solutions.

NET BUSINESS SOLUTIONS (IDFC FIRST BANK) | Mumbai, India

Jun 2020 – Jul 2023

Lead Data Analyst – Internal Audit Analytics

- **Designed and automated data extraction pipelines** using SQL and PySpark from HDFS/Cloudera, processing large-scale banking datasets across multiple CBS platforms to support internal audit cycles.
- **Built Matplotlib and Power BI visualizations** for audit findings and risk trend analysis, enabling the audit committee to identify control gaps 30% faster than prior manual processes.
- **Spearheaded Fintech solution development** and delivery of complex trigger audit reports, integrating data from 4+ core banking systems to provide a unified audit intelligence layer.

- **Implemented SAS Job Execution workflows** and Jupyter Notebook-based automation solutions, reducing ad-hoc analytics turnaround time and standardizing audit data processing.
- **Collaborated with cross-functional teams** to validate ML model outputs and translate results into actionable audit recommendations, directly influencing risk remediation strategies.

C-EDGE TECHNOLOGIES (TCS-SBI ENTERPRISE) | Mumbai, India

May 2018 – Jun 2020

Assistant System Analyst – Core Banking & Data Operations

- **Optimized critical data extraction SQL queries** on Oracle 11g/12c using indexing, constraints, and advanced joins, reducing query execution time and improving data pipeline reliability for RRB and NABARD banks.
- **Managed day-to-day CBS operations** for B@NCS24 including H2H, NEFT, and RTGS payment processing, ensuring zero-downtime SLA compliance across 100+ bank branches.
- **Led data restoration and incident management** for application data handling failures, reducing recovery time through documented runbooks and proactive log monitoring.
- **Developed Unix shell scripts** to automate application availability checks and data processing verification, eliminating manual monitoring overhead.
- **Conducted server space management analysis** across application infrastructure to maintain system stability and performance continuity.

EDUCATION

B.E. Computer Engineering

2017

Rizvi College of Engineering, Mumbai University

CERTIFICATIONS & ACHIEVEMENTS

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- **Master Data Science Program** – GUVI, IIT Madras | Advanced data science, ML, and analytics
 - **AWS Machine Learning** – Knowledge Academy | Cloud-based ML deployment and model management
 - **Deep Learning A-Z** – Hands-On Artificial Neural Networks in Python (Udemy)
 - **Python & Web Technologies Level I** – Suven Consultancy
 - **Google Science Fair Participant (Under 18)** – Robotics hand using inter-kinematics technique with Arduino
 - **Runner-Up, Counter-Strike Tournament** – Rizvi College Technical Fest 2015
 - **Transform Maharashtra Contestant** – Entrepreneurship and innovation challenge